

CSE563 — Assignment 5

Unit V: Algorithm Complexity Analysis

Maps to **CO5** (determine the algorithmic complexity of various mathematical problems). **Marks: 50.** Attempt all questions. Justify every complexity claim — a bare $O(\cdot)$ answer earns half marks at most.

Part A — Short Answer and Definitions (10 marks, 2 each)

Q1. [2 marks] Define O , Ω and Θ formally. Explain why it is correct but unhelpful to say that binary search is $O(n^2)$.

Q2. [2 marks] Explain the difference between “worst case” and “ O ”. Give an algorithm whose best and worst cases have different orders.

Q3. [2 marks] State the three cases of the master theorem. Give one recurrence to which it does not apply and say why.

Q4. [2 marks] Define the classes P and NP. What exactly would it take to prove $P = NP$?

Q5. [2 marks] State the two steps required to prove a problem X is NP-complete, and name the problem proved NP-complete first and by whom.

Part B — Complexity Analysis and Recurrences (24 marks)

Q6. [5 marks] Determine the Θ complexity of each of the five fragments below, with justification.

(a)

```
1 s = 0
2 for i in range(n):
3     for j in range(n):
4         s += i*j
```

(b)

```
1 i = n
2 while i > 1:
3     i = i // 3
```

(c)

```
1 for i in range(n):
2     j = 1
3     while j < n:
4         j *= 2
```

(d) for i in range(n): for j in range(i, n): $O(1)$ body

(e) $T(n) = 4n^3 + 100n^2 \log n + 5^{10}$

Q7. [5 marks] Prove from the definition that $4n^2 + 10n + 7 = \Theta(n^2)$, giving explicit constants c_1, c_2, n_0 . Then prove that $2n^3 \neq O(n^2)$.

Q8. [6 marks] Solve each recurrence, stating the method used. Assume $T(1) = 1$. (a) $T(n) = T(n-1) + n^2$ (b) $T(n) = 2T(n/2) + n^2$ (c) $T(n) = 3T(n/3) + n$ (d) $T(n) = T(n/2) + n$ (e) $T(n) = 4T(n/2) + n^2$ (f) $T(n) = 2T(n-1) + 1$

Q9. [4 marks] Solve the linear recurrence $a_n = 6a_{n-1} - 8a_{n-2}$ with $a_0 = 2$, $a_1 = 10$, using the characteristic-equation method. Verify a_2 two ways.

Q10. [4 marks] Use a recursion tree to solve $T(n) = 2T(n/2) + n$, then prove your answer by the substitution method (induction).

Part C — Complexity Classes, SAT, Searching and Sorting (16 marks)

Q11. [5 marks] Consider $\Phi = (x_1 \vee \neg x_2) \wedge (\neg x_1 \vee x_3) \wedge (\neg x_2 \vee \neg x_3) \wedge (x_2 \vee x_3)$. (a) Determine whether Φ is satisfiable; if so give a satisfying assignment and verify it clause by clause. (b) How many assignments would exhaustive search examine in the worst case for n variables? (c) Name the class Φ 's decision problem belongs to, given that every clause has at most 2 literals.

Q12. [5 marks] $A = [4, 8, 15, 16, 23, 42, 57, 61, 78]$ (sorted, indices 0–8). (a) Trace binary search for the key 23, showing lo, hi, mid and the comparison at each step. (b) Trace binary search for the key 30 and state the result. (c) State the maximum number of comparisons for $n = 9$ and for $n = 10^6$, and justify with the decision-tree lower bound.

Q13. [6 marks] (a) Trace one Lomuto partition of $A = [6, 3, 9, 2, 8, 5]$ with the last element as pivot, showing the array after each swap and the final pivot index. (b) State the best- and worst-case complexity of quicksort and describe the input that triggers the worst case here. (c) Explain why comparison-based sorting cannot beat $\Omega(n \log n)$, and name one sorting algorithm that does run in linear time, stating the assumption that permits it.